

Nexus Librarian MCP v2.5

The Sovereign Developer Memory & Coaching System

1. Executive Summary

The **Nexus Librarian** is a specialized Model Context Protocol (MCP) server acting as a "Sovereign Developer Memory." It bridges the gap between static code repositories and active developer intelligence by integrating local LLM inference (Gemma 4) with persistent vector storage. The system provides semantic history, architectural coaching, and persona-aligned code generation exclusively within the user's local environment.

2. Technical Stack & Hardware

AI Brain: Local Gemma 4 (26B MoE / 31B Dense) via <i>Ollama</i> .	MCP Framework: <i>fastmcp</i> (Python SDK).
Vector Database: <i>ChromaDB</i> (Persistent).	Interface: <i>Streamlit</i> (Command Center UI).
Ingest: <i>GitPython</i> & <i>PyGithub</i> .	Hardware: <i>Apple Silicon M4 Max</i> (Unified Memory).

3. Functional Toolset

Tool	Parameters	Core Logic / Requirement
sync_repository	repo_url, branch	Incremental sync via Watchdog. Uses Delta Tracking ($\Delta = \text{current} - \text{previous}$) to re-index only modified files.
query_memory	query, tech_filter	

Tool	Parameters	Core Logic / Requirement
		Semantic search (e.g. "Where have I used Pydantic?") returned with intent-summarized snippets.
get_dev_persona	scope	Synthesizes extracted "Tendencies" into a stylistic developer profile (Persona).
analyze_diff	diff_text	Gemma 4 explains the "Why" behind code changes relative to the historical persona.
generate_skills_pkg	target_path	ID: FR-SKILLS-01. Exports a PROJECT_SKILLS.md file summarizing the repo's "Professional DNA."

4. Intelligence Layer & Logic

The Ingest Loop



The Tendency Engine

- **Pattern Extraction:** Analysis of code blocks to identify preferences (e.g., Pydantic vs. Dataclasses).
- **AI Coaching:** Proactive alerts when new code deviates from established project patterns.
- **Multi-modal Parsing:** Unified ingestion of .md, .pdf, and source code.

5. Data Strategy (ChromaDB)

Field	Type	Function
content_hash	String	SHA-256 for idempotent indexing and duplicate detection.
tech_stack	List	Literal tags (e.g. ["FastAPI", "Qdrant"]) for filtered search.

Field	Type	Function
intent_summary	Text	The AI-generated "Rationale" behind a specific implementation.
semantic_type	String	Category: Logic, Config, Boilerplate, or Interface.
author_identity	String	Maps patterns to the specific developer's "Tendency Profile."

6. Sovereignty & Performance

Zero-Trust Architecture: No data leaves the local M4 Max. All embeddings and inference are processed via Ollama. Middleware ensures PII and internal IPs are masked before the "Librarian" indexes the content.

- **Throughput:** Leveraging M4 Max UMA for *> 50 tokens/sec* during summarization.
- **Parallelism:** File parsing utilizes all performance cores for low-latency indexing.